Unit 2 : 2 marks

1. What is meant by data encapsulation.
2. Differentiate between Procedure-Oriented Programming and Object-Oriented Programming with respect to data and program organization.
3. Examine the purpose of cin and cout statements in C++. Give one example for each.
4. Distinguish between a class and an object in C++ with a suitable example.
5. Distinguish between a structure and a class in C++.
6. Discuss how private access specifiers differ from protected Access specifiers in C++.
7. Identify the role of member functions in a C++ class. Give a suitable example.
8. Use the Scope Resolution Operator (::) for defining a member function outside a class.
9. Explain the concept of a friend function in C++. Why is it used?
10. Explain the concept of a friend class in C++. What access does a friend class get?
11. Explain the purpose of an inline function in C++. Mention one advantage and one limitation.
12. Explain the concept of a static data member. How is it different from an ordinary data member?
13. Explain the purpose of a static member function in C++. How is it accessed?
14. Explain this pointer in C++. What does it represent?
15. Explain the concept of pointers in C++. What are the uses of the address-of (&) and dereference (*) operators?
16. Explain the difference between call by value and call by reference in C++.
17. Explain how an object can be passed as an argument to a function in C++.
18. Explain how a function can return an object in C++. Give a suitable example.
19. Explain object assignment in C++. What happens when one object is assigned to another object of the same class?
20. Explain how static data members and static member functions are related in C++. Give a suitable example.

Essay Questions:

1.Analyze the major features of Object-Oriented Programming with suitable examples.

2. Distinguish between Procedure-Oriented Programming and Object-Oriented Programming.

3. Write a C++ Program to develop a Student class with data members rollno, name, and marks in three subjects, and methods getdata() and displaydata(). Create two objects St1 & St2 and display the percentage of marks secured by each student.

4.Write a C++ Program having a Bankaccount class with data members account_num, Accountholder_name, and balance. The methods as deposit(), withdraw(), and displaybalance(). Create three objects and demonstrate their use.

5.Write a C++ program to create structure and a class for an employee to read theGive suitable examples. Employee_id & Employee_name and display the details using the structure & using the class.

6. Explain the access specifiers private, public, and protected in C++ with suitable examples.

7. Explain member functions in C++. Discuss their declaration, definition, and invocation with a suitable example.

8. Write a C++ program to apply the Scope Resolution Operator (::) by defining the member methods of a Rectangle class outside the class definition. Use data members length and breadth and methods calculateArea() and display().

9.Write a C++ program using the concept of friend functions to add two distance objects using a distance class with data members feet and inches and display the result.

10. Explain friend classes in C++. Describe how a friend class can access the private and protected members of another class.

11. Write a C++ program to demonstrate the use of an inline function. Define an Employee class with data members basic and allowance, and a member method Gross_salary(). Use an inline function to calculate and display the gross salary.

12. Define an Employee class with data members empId and name, and a static data member count. Create multiple objects and use suitable member methods to increment and display the total number of employees created.

13. Explain static member functions in C++. Discuss their characteristics and explain how they differ from ordinary member functions.

13. Explain this pointer in C++. Discuss its purpose and illustrate its use with a suitable example.
14. Explain the concept of pointers in C++. Discuss pointer declaration, initialization, dereferencing, and pointer arithmetic with examples.
15.Write a C++ program using a Student class with data members rollno and marks, and methods getdata() and display(), to demonstrate call by value & call by reference techniques.
16. Write a C++ program using an Employee class with data members empid and salary, and methods getdata() and display(). Pass an object to a function and display its details.
17. Write a C++ program using a Rectangle class with data members length and breadth, and methods getdata(),Area(), and display(). Pass a Rectangle object to a function.
18. Write a program using a Distance class with data members feet and inches, and methods getdata(), add(), and display(), where a function returns a Distance object.
19. Write a C++ program using a Book class with data members bookId, title, and price, and methods getdata() and display(), and demonstrate object assignment.
20. Write a C++ program using an Employee class with data members Empno and Empname, and methods getdata() and display() to demonstrate the concept of object assignment and passing objects to functions.

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